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Systematic Direct Observation of Time on Task as a Measure of Student Engagement

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This study examines the relationship between systematic direct observation (SDO) of time on task (TOT) and student engagement as measured by a self-report task-specific measure. The research questions guiding the study are (a) what is the relationship between SDO of TOT and a self-report measure of student effort, and (b) what is the relationship between TOT and reading comprehension? An exploratory analysis compares the engagement among three levels of passage difficulty. Small partial correlations are found between TOT and self-reported engagement among 125 third and fourth graders ($r = -.15$ and $r = .30$, respectively). Nonsignificant partial correlations between TOT and reading comprehension are also found. Exploratory multivariate analyses of the three variables show significant results for both third and fourth grade, but subsequent univariate analyses find a significant effect only for comprehension. Thus, these data suggest the need for additional research regarding the assessment of TOT with SDO.

Keywords: *observation; time on task; engagement*

Academic learning time (ALT) is defined as the "proportion of instructional time allocated to a content area during which students are actively and productively engaged in learning" (Gettinger & Seibert, 2002, p. 774) and is closely linked to academic outcomes (Fisher & Berliner, 1985; Gettinger & Stoiber, 1999). ALT is comprised of two components, procedural engagement and substantive engagement (Nystrand & Gamoran, 1991). Procedural engagement is the observable behavior associated with learning such as completing tasks (Gettinger & Seibert, 2002), whereas substantive engagement is the "sustained commitment to and engagement in the content of schooling" (Nystrand & Gamoran, 1991, p. 262). Thus, substantive engagement is the student's active involvement in the learning task (Reeve, Jang, Carrell, Jeon, & Barch, 2004), which, for example, in reading is often measured as the number of words read or amount of text comprehended (Appleton, Christenson, Kim, & Reschly, 2006). Intervention research tends to focus on procedural engagement because of the two components, it is the more observable and can be more easily manipulated (Albers, Elliott, Kettler, & Roach, 2005), but substantive engagement represents the students' investment in learning, which directly affects effort and persistence (Fredricks, Blumenfeld, & Paris, 2004).

Because of its importance to learning, substantive engagement has been measured in many ways, all of which are related to student outcomes. Quality of instructional discourse in the classroom was positively related to a measure of students' in-depth understanding (Nystrand & Gamoran, 1991), and measures of persistence and the use of effort management strategies were associated with increased academic performance on a variety of tasks (Corno, 1986; Miller, Greene, Montalvo, Ravindran, & Nichols, 1996; Pintrich & DeGroot, 1990). Moreover, other variables such as grade point average, credits accrued in middle and high school, attendance, classroom participation, and participation in extracurricular activities have been correlated with engagement and are often used as indicators of engagement (Appleton et al., 2006).

Assessing substantive engagement in the learning process is potentially useful but is difficult to do (Fredricks et al., 2004; Nystrand & Gamoran, 1991; Pintrich, Wolters, & Baxter, 2000; Winne & Perry, 2000). Because procedural engagement is more observable than substantive engagement, it involves fewer inferences and

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can be observed within the context of a classroom (Skinner, Rhymer, & McDaniel, 2000). Greenwood, Horton, and Utley (2002) assessed student engagement with classroom observations of academic responding (responding to situations, commands, and instructions) and task management (behaviors in which students prepare to respond). Many studies use the orienting of student behavior (e.g., having his or her head oriented toward the appropriate instructional presentation medium) as an operational definition of observable engagement, but orientation of view can be misleading (Skinner et al., 2000). Thus, Shapiro (2004) suggests assessing active engagement with on-task behaviors, which are defined as the student "attending to his/her work or assigned task/activity" (p. 99), such as writing, raising his or her hand, oral reading, answering questions, listening to instructions, looking at a worksheet, and so forth. These definitions emphasize procedural engagement as a measure of ALT; however, ALT involves both procedural and substantive engagement, and the appearance of engagement may not actually represent optimal ALT (Gettinger & Seibert, 2002).

Direct observation of student behavior is a common (Wilson & Reschly, 1996) and increasingly prevalent (Reschly, 2003) assessment technique among school psychologists. In systematic direct observation (SDO), the observer precisely and operationally defines the relevant behavior and records the frequency with which the behavior occurs (Salvia, Ysseldyke, & Bolt, 2007). This systematic approach to assess time on task (TOT) led to adequate interrater agreement, but questionable reliability and generalizability (Hintze & Matthews, 2004). Moreover, it is unclear if observable behaviors actually represent the internal behavior of interest (Cone, 1978). Substantive engagement refers to internal activities and qualities of individuals that are not easily observed (Newmann, Wehlage, & Lamborn, 1992). Thus, it is unknown if SDO of TOT actually assesses substantive engagement.

Many variables influence substantive engagement and TOT including interest in the task (Schiefele, 1999), the student's goal for completing the task (Dweck, 1999), and the task difficulty (Burns & Dean, 2005; Gickling & Armstrong, 1978; Treptow, Burns, & McComas, 2007). Of these, task difficulty is the most easily manipulated, and using easier material leads to increases in TOT for children with academic difficulties (DePaepe, Shores, Jack, & Denny, 1996; McComas, Hoch, Paone, & El-Roy, 2000). However, no study could be found that examined the effect of task difficulty on measures of substantive engagement.

Given that substantive engagement must be either inferred from observations of behavior or measured by self-report measures (Fredricks et al., 2004), the purpose of the current study is to examine the relationship between the SDO of TOT, substantive engagement as measured by a self-report task-specific measure, and an assessment of comprehension. The research questions that guided the study were (a) what is the relationship between the SDO of TOT and a self-report measure of student effort and persistence, and (b) what is the relationship between TOT and reading comprehension? Moreover, because no study could be found that examined the effect of task difficulty on measures of substantive engagement, an exploratory analysis was conducted to compare engagement for the three levels of passage difficulty (below grade level, at grade level, and above grade level).

Method

Participants

The participants for the study were 125 students in the third or fourth grade, attending one elementary school in Minnesota. Sixty-one (48.8%) third-grade and 64 (51.2%) fourth-grade students participated, with a relatively equal distribution of females ($n = 61$, 48.8%) and males ($n = 64$, 51.2%). The ethnic breakdown of the sample included 90 (72.0%) Caucasian students, 24 (19.2%) African American students, 9 (7.2%) Hispanic students, and 2 (1.6%) Asian American students. The elementary building served 518 students in kindergarten through fifth grade, with 39.0% eligible for the federal free or reduced lunch program.

Procedures

The principal of the participating school was contacted to obtain initial consent for the study to occur in that building. Next, the third- and fourth-grade general education teachers were contacted, and consent forms were sent home to the parents of children in those rooms. Data were collected in 1 day by five school psychology graduate students with advanced training in assessment. Each also participated in a 3-hr training in SDO as part of course work and received 1 additional hour of training on SDO of TOT. Both of these trainings included direct instruction and facilitated practice. Each participating classroom was observed in a random order. Students were given a reading task to complete while being observed and then were asked to complete a four-item questionnaire regarding their level of effort.

One researcher orally read the student assent form to the entire classroom and asked the students to sign a form granting their assent. Next, the directions were orally read to the students in a group format asking them to read two 1-page sheets and answer eight comprehension questions (four for each page). The students were instructed to silently read each of the two stories one at a time, answer the corresponding questions for the first story, and then complete the second story and questions. The comprehension questions were presented and responded to in writing by selecting the correct multiple choice option.

Students were instructed to turn each sheet over as they finished and quietly work on something else until all students had finished working. The sheets were collected after all students had completed both stories and all comprehension questions. Finally, the students were given a four-item self-report survey of effort as a measure of engagement and were asked to complete the survey and turn it over when finished. The surveys were collected when all students were completed, and the students were thanked for their participation. The data collection sessions ranged from 15 to 25 min in length.

The passages were taken from the Read Naturally (RN) reading series (RN, 2003) in order to control the difficulty level of the passages. RN is a reading program that includes a collection of short reading passages at sequenced levels of reading difficulty and is designed to allow for instruction in and monitoring of students' progress in reading fluency and comprehension. The program includes 24 passages at each of 13 reading levels, ranging from Grade 1 to Grade 8. The passages range in length from 50 to 200 words, with each individual passage followed by five comprehension questions, four of which are multiple choice and one of which is open-ended, but only the four multiple choice questions were used for the study to assure objectivity in scoring procedures. The questions appear on the same page as the reading passage.

Because on-task behavior is correlated with task difficulty (Gickling & Armstrong, 1978), variance within the task difficulty was assured by using three different reading levels. One third of all students received passages that were purportedly grade appropriate (e.g., 3.5 for third grade and 4.5 for fourth grade). One third of the students read a passage that was written at two grade levels below the given grade (e.g., 1.5 for third grade and 2.5 for fourth grade), and one third worked from a passage that was two levels above their grade (e.g., 5.5 for third grade and 6.5 for fourth grade). The Harris-Jacobson (Harris & Jacobson, 1982) Wide Range Readability Formula was used to independently evaluate the readability of the passages. Results suggested readability estimates that were generally consistent with the purported level. Passages used with third-grade students ranged from 1.9 to 5.2, and those used for fourth-grade students ranged from 2.4 to 7.5.

Measures

The three variables of interest to the study were TOT, self-report of student engagement, and comprehension. On-task behavior was defined as actively attending to the assigned instructional material (Shapiro, 2004) and was assessed once for each classroom using a momentary time sampling with 10-s intervals (Hintze, Volpe, & Shapiro, 2002). Examples included looking at the reading material, writing, listening to instructions from the teacher, and raising a hand for assistance from the teacher (Shapiro, 2004). Examples of off-task behavior included talking about anything other than the assigned reading, leaving the seat for nonrelevant reasons, aimlessly moving the reading passages (e.g., flipping back and forth), gazing away from the reading passages, reading something other than the assigned passages, and focusing attention on the activities of others (Shapiro, 2004). The total number of intervals that were judged to be on task was divided by the total number of intervals to convert the data to a percentage of TOT.

Substantive engagement is frequently and perhaps best assessed through self-report measures (Appleton et al., 2006; Fredricks et al., 2004), and student self-monitoring is important for improving ALT (Gettinger & Seibert, 2002). Therefore, a self-report measure of engagement was used for the current study. Student effort and persistence are viewed as critical components of substantive engagement (Fredricks et al., 2004). Therefore, the Effort and Persistence in Learning (EPL) subscale of the *Student Approaches to Learning Survey* (Artelt, Baumert, Julius-McElvany, & Peschar, 2003) was used to assess student self-report of engagement. The subscale consisted of the four items listed in Table 1, which were scored from 1 to 4. Previous research found a coefficient alpha of .83 for the EPL subscale among students in the United States, and the results of the EPL were significantly different between students in the top quartile for reading skills and those in the bottom quartile (effect size = .36; Artelt et al., 2003). Data for the study were the self-report ratings of the four items summed to create a total engaged score, which ranged from 0 to 16.

Comprehension was assessed by counting how many of the four multiple choice comprehension questions the participant answered correctly. The total comprehension score for each student equaled the total answered correctly, out of the eight, for the two passages administered.

Interobserver Agreement

Interobserver agreement (IOA) was computed for TOT by having 25% of the students observed by two independent raters and comparing the percentage of agreement

Table 1
Effort and Persistence Subscale and Frequency of Respondent Endorsement and Mean Scores

When completing the reading assignment I just finished:

	Strongly Disagree		*Disagree*		*Agree*		*Strongly Agree*			
Item	*n*	%	*n*	%	*n*	%	*n*	%	*M*	*SD*
I worked as hard as possible	2	1.6	5	4.1	58	47.2	58	47.2	3.40	0.65
I kept working even if it was hard	1	0.8	6	4.9	46	37.4	70	56.9	3.50	0.63
I tried my best to learn what was taught	0	0.0	10	8.1	52	42.3	61	49.6	3.42	0.63
I put forth my best effort	2	1.6	10	8.1	44	35.8	67	54.5	3.43	0.71

Table 2
Means and Standard Deviations for Comprehension, Time on Task, and Total Engaged Score

	Third Grade ($n = 59$)		Fourth Grade ($n = 64$)		Total ($n = 123$)	
Variable	*M*	*SD*	*M*	*SD*	*M*	*SD*
Comprehension[a]	65.47	21.76	63.18	26.72	64.28	24.34
Time on task[b]	91.36	12.96	96.39	4.20	93.97	9.77
Total engaged score	13.79	1.95	13.70	1.76	13.74	1.85
Time[c]	12.76	4.83	10.59	3.87	11.63	4.47

a. Reported as the percentage of questions answered correctly.
b. Reported as the percentage of 10-s intervals.
c. Reported as the number of minutes required to complete the task.

between the intervals that were rated as on task. In other words, the number of items in which both reviewers rated the behavior as on task or off task were totaled and divided by the total number of intervals. The resulting 93% agreement suggested adequate agreement for purposes of research. Moreover, the number of intervals rated as on task by both observers were corrected with a Pearson product–moment correlation and resulted in a coefficient of .99.

IOA was also computed for the comprehension questions by again having a second observer rate the answers to the comprehension questions as correct or incorrect for 25% of the students. Then the total number of questions in which both raters marked the question as correct or incorrect was divided by the total number of questions. The resulting IOA for this multiple choice task was 100%.

Results

Table 2 lists descriptive statistics for the four variables for each grade group. A multivariate analysis of variance between grades found a significant effect, Wilks's $\Lambda = .88$, $F(4, 119) = 3.87$, $p < .01$, and follow-up univariate analyses found significant effects for TOT, $F(1, 121) = 8.81$, $p < .01$, and time taken to complete the task, $F(1, 121) = 8.04$, $p < .01$. Therefore, analyses carried out to address the research questions were conducted by grade groups.

Time taken to complete the task was correlated with TOT using the Pearson product–moment to assess if the number of intervals needed to complete the task was related to the number of intervals that the children were judged as on task. A significant correlation was found ($r = -.19$, $p < .05$) and suggested a negative relationship between the two variables. Thus, the longer the students required to complete the task, the lower the percentage of TOT. As a result, partial correlations were used to assess the research questions, with time being the covariant.

The first research question addressed the relationship between the SDO of TOT and a self-report measure of student effort and persistence. A partial correlation between TOT and the total engaged score (sum of the items) resulted in nonsignificant correlations for third graders, $r(56) = -.15$, $p = .25$, but a significant correlation for fourth graders, $r(61) = .30$, $p < .05$.

The second research question addressed the relationship between TOT and comprehension. A partial correlation resulted in a nonsignificant correlation between TOT and comprehension for both third-grade students,

Table 3
Means, Standard Deviations, and Analyses of Variance for Variables by Passage Difficulty

	Third Grade							Fourth Grade						
	Below Grade ($n = 20$)		Grade Level ($n = 19$)		Above Grade ($n = 20$)			Below Grade ($n = 21$)		Grade Level ($n = 23$)		Above Grade ($n = 20$)		
Variable	*M*	*SD*	*M*	*SD*	*M*	*SD*	*F*	*M*	*SD*	*M*	*SD*	*M*	*SD*	*F*
Time on task	91.27	17.04	88.16	12.18	94.48	7.85	1.16	95.57	5.03	97.01	3.61	96.39	4.20	0.65
Comprehension	78.00	16.42	60.00	19.44	58.15	23.79	5.90*	77.62	23.00	62.83	28.48	48.43	20.27	7.35*
Total effort	14.00	1.78	13.74	1.91	13.63	2.23	0.19	13.86	1.65	13.33	1.76	13.95	1.88	0.80

Note: Multivariate analysis of variance (MANOVA) for third grade $\Lambda = .77$, $F(6, 110) = 2.49$, $p < .05$. MANOVA for fourth grade $\Lambda = .72$, $F(6, 120) = 3.44$, $p < .05$.
*$p < .01$.

$r(56) = .15$, $p = .26$, and fourth-grade students, $r(61) = .23$, $p = .07$.

Finally, an exploratory multivariate analysis of the three variables according to the passage difficulty level was conducted using below grade level, at grade level, and above grade level as the three levels of the grouping variable. A significant effect was noted for both the third-grade students, Wilks's $\Lambda = .77$, $F(6, 110) = 2.49$, $p < .05$, and the fourth-grade students, Wilks's $\Lambda = .72$, $F(6, 120) = 3.44$, $p < .01$. However, as also can be seen in Table 3, the subsequent univariate analyses found a significant effect only for comprehension among both grade groups. Thus, the difficulty level of the passage seemed to affect reading comprehension, but not TOT or self-reported persistence and effort.

Discussion

The first research question inquired about the relationship between SDO of TOT and a self-report of student effort and persistence. The coefficients represented small relationships, but a significant coefficient was noted for fourth and not for third grade. There are several potential interpretations including developmental appropriateness of the data. The older a student is, the more reliable self-report data are (Kamphaus & Frick, 2005). It seems improbable that a 1-year age difference could result in such substantial disparity, but the results are consistent with previous research (Kamphaus & Frick, 2005). Moreover, students in fourth grade were on task for more than 96% of the intervals. Thus, those students may have been more interested in presenting themselves in a more positive light. However, it could also be that TOT is a more meaningful construct for fourth-grade children and that they are more accustomed to completing academic tasks that are consistent with definitions of TOT. These are hypotheses in need of additional research.

It is difficult to interpret correlations between two measures with interpretive rules, but coefficients of .30 to .40 are common (Kaplan & Saccuzzo, 2001). The coefficients noted above suggest a weak relationship between SDO of TOT and self-reported task-specific engagement. In fact, the coefficients are probably too low to support use of one as a measure of the other, especially with third graders. Future researchers may wish to replicate this study with older children or develop a different measure of task engagement that does not rely on self-report methods.

Although the second research question addressed the relationship between TOT and reading comprehension, the magnitude of relationship was similar to the magnitude between TOT and engagement. Again, the correlation was not significant for third graders, but was for fourth graders. It makes intuitive sense that persistence and effort in engagement would lead to better reading comprehension. Previous research found that mild time pressure increased reading comprehension among adults, probably due to enhanced attention to the task (Walczyk, Kelly, Meche, & Braud, 1999). The current data question the relationship between SDO of TOT and academic performance, or at least the measurement thereof. The students demonstrated relatively low comprehension by correctly answering only 64.28% of the questions. This low level of comprehension could question the effort applied by students toward this academic task, but previous research found similar average percentages of comprehension questions correct among average adult and child readers (45%, Lagrou, Burns, Mizerek, & Mosack, 2006; 49.4% and 70.7%, Pressley, Ghatala, Woloshyn, & Pirie, 1990; 65.31% to 74.86%, Walczyk et al., 1999).

The current data question the use of TOT as a measure of task engagement or academic behavior for this sample. SDO remains a required component of special education eligibility assessments and is becoming more frequently used among school psychologists (Reschly, 2003). Although SDO is used for constructs other than TOT, assessment of TOT is a common behavioral target of SDO. Therefore, these data could be considered in combination with the reliability estimates found by Hintze and Matthews (2004) when considering using SDO of TOT for educational decisions, especially when making important decisions about individuals such as special education eligibility, which requires that the highest psychometric standard be met (Salvia et al., 2007).

Previous research found that task difficulty was related to TOT (Burns & Dean, 2005; Gickling & Armstrong, 1978; Treptow et al., 2007). However, only comprehension was significantly affected by task difficulty; TOT was not. It makes intuitive sense that a child would demonstrate lower comprehension of more difficult passages, but these data are inconsistent with the aforementioned previous studies. However, the previous studies measured reading difficulty with percentage of words known within the reading passage for each child. The current study relied on estimates of readability, which have questionable validity (Ardoin, Suldo, Witt, Aldrich, & McDonald, 2005). Thus, future researchers could examine this question further with more precise estimates of reading difficulty.

As is often the case with research, the current data create more questions than they answer. The low correlation coefficients question the relationship between SDO of TOT and engagement or academic performance. Thus, what exactly is measured by an SDO of TOT is somewhat unknown, and this study could be viewed as only the first in a line of necessary research. In addition to suggestions mentioned above, future researchers could examine the relationship between engagement and work products or product completion. Moreover, a more precise examination of reading tasks than percentage of comprehension questions answered correctly (e.g., oral reading fluency) could also be used to examine engagement. Substantive engagement is important and is consistently linked to positive academic outcomes (Nystrand & Gamoran, 1991), but it remains a problematic construct to assess and warrants additional research.

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